

School of Computing

MSc Project Report

Sentiment Analysis and Sarcasm Detection in Product
Reviews using NLP and LLMs

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This report is submitted as part of the requirements for the degree of

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I confirm that the work contained in this Honours project report has been composed solely by myself and has not been accepted in any previous application for a degree. All sources of information have been specifically acknowledged and all verbatim extracts are distinguished by quotation marks.

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1 ABSTRACT

This study explores sentiment analysis and sarcasm detection in online product reviews using different natural language processing approaches. With the increasing use of e-commerce platforms, analysing customer feedback has become important for understanding user satisfaction. However, one major challenge is sarcasm, where users express negative opinions using positive words, making it difficult for automated systems to correctly interpret the true meaning of the text.

In this project, multiple models were implemented and compared, including a traditional machine learning model (Support Vector Machine), transformer-based models (fine-tuned BERT and zero-shot BERT), and a large language model (LLaMA-3). The dataset was prepared by extracting review text and ratings, which were then used for sentiment classification and sarcasm detection tasks.

The results show that the fine-tuned BERT model achieved the highest overall accuracy across both tasks, while the SVM model performed well on structured data but relied heavily on word patterns. The zero-shot BERT and LLaMA-3 models demonstrated better generalisation on unseen examples, particularly in detecting sarcasm, but showed lower consistency in sentiment classification.

The findings highlight that while advanced models improve contextual understanding, sarcasm remains a challenging problem in sentiment analysis. The study also shows that there is a trade-off between accuracy and generalisation when comparing different models.

Overall, this work provides a comparative analysis of different approaches and highlights the importance of combining sentiment analysis with sarcasm detection for more accurate interpretation of customer reviews.

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2 REPORT STRUCTURE

This report is organised into seven chapters.

Chapter 1 introduces the research topic, including the background, problem statement, aim, objectives, research questions, motivation, and scope of the study.

Chapter 2 reviews existing literature on sentiment analysis, sarcasm detection, emotion understanding, and large language models. It also identifies the research gap addressed in this study.

Chapter 3 presents the methodology used in the project. It explains the system workflow, dataset preparation, preprocessing steps, model selection, evaluation metrics, tools, and validation strategy.

Chapter 4 describes the implementation of the system. It explains how the dataset was prepared, how preprocessing was applied, and how each model was implemented, including SVM, fine-tuned BERT, zero-shot BERT, and LLaMA-3.

Chapter 5 presents the testing and results of the implemented models. It reports model performance for both sentiment analysis and sarcasm detection.

Chapter 6 provides evaluation and discussion of the results. It analyses model behaviour, word importance, generalisation, limitations, and comparative performance.

Chapter 7 concludes the study and presents future work, including improvements such as real-world sarcastic review collection, emotion detection, and combined sarcasm-sentiment analysis.

GANTT CHART

The following Gantt chart illustrates the timeline and key phases of the project.

	Jan29	Feb	Mar	Apr25
Testing and Evolution				
Model Implementation (LLaMA-3)				
Model Implementation (BERT)				
Model Implementation (SVM)				
Data Preprocessing				
Data collection				

Fig x: Gantt chart showing the timeline of project activities

3 CHAPTER 1: INTRODUCTION

3.1 Background and Importance

In recent years, online shopping has become a major part of everyday life. Customers often rely on product reviews to make decisions before purchasing items. These reviews provide valuable insights into product quality, performance, and user satisfaction. As a result, analysing customer feedback has become important for both businesses and consumers.

Sentiment analysis is widely used to understand whether a review expresses a positive, negative, or neutral opinion. It helps companies identify customer satisfaction levels and improve their products. However, one major challenge in analysing reviews is the presence of sarcasm. In many cases, users express negative opinions using positive words, which can mislead automated systems [1], [7].

For example, a review such as *“Great product, stopped working in one day”* may appear positive at first but actually conveys dissatisfaction. This makes it difficult for traditional models to correctly interpret the true meaning of the text. As a result, sarcasm detection has become an important area of research in natural language processing [1].

Understanding sarcasm is essential for improving the accuracy of sentiment analysis systems. Without it, systems may produce incorrect results, leading to poor decision-making. Therefore, combining sentiment analysis with sarcasm detection can provide a more accurate understanding of user feedback in online product reviews [4].

3.2 Problem Statement

Although sentiment analysis has improved significantly with the development of machine learning and deep learning models, it still faces challenges when dealing with complex language patterns such as sarcasm. Most traditional models rely on word-level patterns and may fail to capture the true meaning when there is a contradiction between the words used and the intended message [1], [7].

In product reviews, sarcasm is often used to express dissatisfaction in an indirect way. This creates confusion for automated systems, as the literal meaning of the sentence does not match the actual opinion of the user. As a result, models may incorrectly classify sarcastic reviews as positive, reducing the overall reliability of sentiment analysis systems [4].

Furthermore, many existing approaches treat sentiment analysis and sarcasm detection as separate tasks. This limits the ability to fully understand user intent, as sarcasm often changes the underlying sentiment of a review. There is also a lack of focus on how different types of models, such as traditional machine learning models, transformer-based models, and large language models, perform on these tasks [6], [8].

Therefore, there is a need to explore and compare different approaches to better understand how sarcasm affects sentiment analysis and how model performance can be improved.

3.3 Research Aim and Objectives

Aim

The aim of this study is to analyse and compare different models for sentiment analysis and sarcasm detection in product reviews, and to understand how these models handle complex language patterns such as sarcasm.

Objectives

- To preprocess and prepare a dataset of product reviews for sentiment analysis and sarcasm detection
- To implement a traditional machine learning model (SVM) for classification tasks
- To implement transformer-based models such as fine-tuned BERT and zero-shot BERT
- To evaluate the performance of a large language model (LLaMA-3) using prompt-based inference
- To compare the performance of different models based on accuracy and generalisation
- To analyse how sarcasm affects sentiment classification

3.4 Research Questions

This study aims to answer the following research questions:

- How do different models perform in sentiment analysis and sarcasm detection tasks?
- How does sarcasm affect the accuracy of sentiment classification?
- Do large language models perform better than traditional models in understanding complex language?
- How well do zero-shot models generalise to unseen data?

3.5 Motivation

The motivation for this study comes from both academic interest and personal experience. During online shopping, I encountered a situation where I purchased a product that turned out to be defective. My immediate reaction was to describe it sarcastically, saying *“Wow, what a wonderful product”*, even though I was actually very dissatisfied.

This experience highlights how common sarcasm is in real-world reviews and how it can easily mislead automated systems. A model that only looks at the words *“wonderful product”* may incorrectly classify such a review as positive, even though the true sentiment is negative.

This real-life scenario motivated the need to explore how different models interpret such language and whether they can correctly understand the intended meaning behind sarcastic expressions. It also emphasises the importance of improving sentiment analysis systems so that they reflect real user opinions more accurately.

3.6 Scope of the Study

This study focuses on analysing sentiment and sarcasm in online product reviews using natural language processing techniques. The scope of the project is limited to text-based analysis of customer reviews obtained from a publicly available dataset.

The study includes the implementation and comparison of different types of models, including a traditional machine learning model (SVM), transformer-based models (fine-tuned BERT and zero-shot BERT), and a large language model (LLaMA-3). These models are evaluated based on their ability to classify sentiment and detect sarcasm in product reviews.

The project focuses only on English-language text and does not include multilingual analysis. In addition, the study does not involve real-time data processing or deployment of the models in a production environment. The sarcasm dataset used in this project includes both real and generated reviews, and the analysis is limited to this dataset.

While emotion understanding is discussed as part of the research context, a separate emotion classification model is not implemented in this study. Instead, the focus is on understanding how sarcasm affects sentiment interpretation.

Overall, the scope of this study is to evaluate how different models perform in handling sentiment and sarcasm, rather than building a complete real-world system.

4 Chapter 2: LITERATURE REVIEW

4.1 Sentiment Analysis in Online Product Reviews

Sentiment analysis, commonly referred to as opinion mining, focuses on identifying the polarity expressed in textual data, typically categorised as positive, negative, or neutral. In the context of e-commerce, sentiment analysis plays an important role in extracting insights from customer reviews, enabling organisations to understand user satisfaction, identify product issues, and support decision-making processes. Early approaches to sentiment analysis relied heavily on lexicon-based techniques and traditional machine learning models such as Naïve Bayes, Logistic Regression, and Support Vector Machines. These methods depend largely on predefined sentiment dictionaries or manually engineered features, which limits their ability to capture contextual nuances in language [6].

With the advancement of deep learning, models such as Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and transformer-based architectures have significantly improved performance by modelling contextual dependencies within text. Transformer models, in particular, introduced attention mechanisms that allow the model to consider relationships between all words in a sentence simultaneously, thereby improving contextual understanding. Despite these improvements, sentiment analysis systems still face challenges when the literal meaning of the text differs from the intended meaning, as is often the case with sarcasm [6].

In product reviews, this limitation becomes particularly evident. Reviews frequently contain implicit expressions, exaggeration, or irony, where the surface-level sentiment does not align with actual user experience. As a result, even advanced models can misclassify sentiment when relying solely on lexical cues without deeper contextual reasoning.

4.2 Sarcasm as a Challenge in Sentiment Analysis

Sarcasm is a form of figurative language in which the intended meaning contrasts with the literal wording. It often involves expressing negative sentiment using seemingly positive language or vice versa. This characteristic makes sarcasm particularly challenging for computational models, as it requires understanding not only the words used but also the underlying intent and contextual indicators.

A widely cited survey on sarcasm detection highlights that sarcasm frequently involves features such as polarity contrast, contextual incongruity, emotional contradiction, and pragmatic features [1]. For example, a statement such as *“Great product, stopped working after one use”* contains a positive lexical indicator (“great”) but conveys dissatisfaction through contextual contradiction. Traditional sentiment analysis models tend to focus on individual word polarity and therefore may incorrectly classify such statements as positive.

Recent research continues to emphasise that sarcasm detection is inherently complex because it requires integrating multiple sources of information, including linguistic patterns, contextual relationships, and emotional signals [4], [8]. In addition, sarcasm often depends on implicit knowledge or real-world expectations, which further complicates automatic detection.

In the context of online product reviews, sarcasm is commonly used by customers to express frustration or disappointment in a subtle or humorous manner. This makes it difficult to accurately interpret user feedback, as the true sentiment may be masked by seemingly positive expressions.

4.3 Traditional and Deep Learning Approaches to Sarcasm Detection

Before the emergence of large language models, sarcasm detection was addressed using a variety of approaches. Early methods relied on rule-based systems and handcrafted features, such as punctuation patterns, intensifiers, sentiment shifts, and lexical contrasts [1]. These approaches were limited in their ability to generalise, as they depended heavily on manually defined rules.

Machine learning models such as Support Vector Machines and Random Forests were later introduced to learn patterns from annotated datasets. These models improved performance but still required extensive feature engineering. The development of deep learning techniques marked a significant shift in sarcasm detection, enabling models to automatically learn hierarchical representations of text.

Neural architectures such as CNNs and LSTMs were used to capture local and sequential patterns, while attention-based models improved the ability to focus on relevant parts of the text. More recently, transformer-based models have become the dominant approach due to their superior ability to capture long-range dependencies and contextual relationships.

For instance, Vitman et al. proposed a sarcasm detection framework that integrates contextual, sentiment, and emotion-based features using transformer models [4]. Their findings demonstrated that incorporating emotional and contextual information can significantly improve sarcasm detection performance. Similarly, Jang et al. highlight that although modern models perform well on benchmark datasets, they often struggle to generalise across different domains and datasets [8]. This suggests that sarcasm detection remains an open challenge despite advances in model architectures.

4.4 Dataset Challenges in Sarcasm Detection

The effectiveness of sarcasm detection models is strongly influenced by the quality and design of the dataset used for training and evaluation. Many early datasets were created using distant supervision

techniques, such as collecting social media posts containing hashtags like #sarcasm. While this approach allows for large-scale data collection, it often introduces noise and bias, as the presence of the hashtag does not always guarantee genuine sarcasm.

To address these limitations, more recent datasets have focused on capturing author-intended sarcasm. The SemEval-2022 iSarcasmEval dataset is a notable example, as it includes annotations based on the author's intended meaning rather than external interpretation [3]. This reduces ambiguity and improves the reliability of labels. The dataset also includes non-sarcastic rephrasing's, which provide valuable context for understanding how meaning changes when sarcasm is removed.

However, challenges remain. Sarcasm datasets are often domain-specific, and models trained on one dataset may not perform well on another due to differences in language style, context, and annotation methods. Additionally, synthetic or artificially generated datasets, while useful for experimentation, may lack the variability and subtlety found in real-world data.

These challenges highlight the importance of carefully designing datasets and interpreting model performance with caution, particularly when evaluating generalisation capabilities.

4.5 Emotion Understanding in Text

While sentiment analysis focuses on identifying the overall polarity of a text, emotion understanding aims to capture more fine-grained affective states such as anger, frustration, disappointment, and joy. This provides a deeper insight into user experiences, as reviews with the same sentiment can express very different emotional responses. For instance, a negative review may reflect mild dissatisfaction, strong frustration, or even sarcasm, each of which carries different implications for product evaluation.

Recent work in this area has highlighted the importance of moving beyond simply polarity classification. The GoEmotions datasets introduced by Demszky et al. demonstrated that human emotions in text are diverse and often overlap, requiring more detailed modelling than traditional sentiment categories [2]. This is particularly relevant in user-generated content, where emotional expression tends to be informal, implicit, and context-dependent.

However, accurately identifying emotions becomes significantly more challenging in the presence of sarcasm. Sarcastic language often masks the true emotional intent of the writer by using positive wording to convey negative feelings. For example, a review such as *"Absolutely brilliant-it stopped working immediately"* may appear positive at a lexical level, while actually expressing frustration or disappointment. As a result, models that rely solely on surface-level sentiment patterns may fail to capture the underlying emotional meaning.

While emotion detection can provide valuable insights, it is not always necessary to implement a separate emotion classification model in order to study emotional understanding. In this project, emotion is examined indirectly through the interaction between sentiment analysis and sarcasm detection. By identifying when sentiment is reversed or distorted due to sarcasm, it becomes possible to infer the underlying emotional intent more accurately.

Therefore, this study focuses on improving the interpretation of customer emotions by combining sentiment analysis with sarcasm detection, rather than performing standalone emotion classification. The approach allows for a more practical and focused investigation while still addressing the broader goal of understanding emotional expression in sarcastic product reviews.

4.6 Large Language Models for Sentiment Analysis

The introduction of large language models (LLMs) has significantly transformed natural language processing. These models are pre-trained on large corpora using self-supervised learning, allowing them to capture rich semantic and contextual information. One of the key advantages of LLMs is their ability to perform tasks using zero-shot and few-shot prompting, reducing the need for large labelled datasets.

Recent studies have shown that LLMs can achieve strong performance in sentiment analysis tasks, particularly in scenarios involving complex language structures and contextual dependencies. However, their performance is not always consistently superior to smaller, task-specific models. Zhang et al. found that while LLMs perform well on general sentiment tasks, they do not always outperform fine-tuned models on specialised datasets [6].

This suggests that although LLMs offer flexibility and generalisation capabilities, their effectiveness depends on the nature of the task and the evaluation setting. Therefore, it is important to consider both traditional and LLM-based approaches when analysing sentiment in complex scenarios such as sarcasm.

4.7 Large Language Models for Sarcasm Detection

LLMs are particularly well-suited for sarcasm detection due to their ability to model contextual relationships and infer implicit meaning. Unlike earlier models that rely on explicit feature engineering, LLMs can learn patterns contradiction, expectation, and emotional reversal directly from data.

Recent research has explored how prompting strategies can enhance sarcasm detection using LLMs. Sun et al. demonstrate that prompting LLMs to identify relevant contextual signals before classification can improve performance in text classification tasks [5]. This approach is particularly relevant for sarcasm detection, as it requires the model to consider multiple linguistic and contextual factors simultaneously.

Similarly, Sun et al. demonstrate that prompting LLMs to identify relevant clues before classification can improve performance in text classification task [5]. This approach is particularly relevant for sarcasm detection, as it requires the model to consider multiple linguistic and contextual factors simultaneously.

In addition, Jang et al. emphasise that generalisation remains a key challenge in sarcasm detection, as models often perform well on specific datasets but fail to generalise across domains [8]. These findings suggest that the effectiveness of LLMs in sarcasm detection depends not only on the model itself but also on how it is prompted and evaluated.

4.8 Research Gap

Although significant progress has been made in sentiment analysis, sarcasm detection, and emotion classification, these tasks are often studied independently. Traditional sentiment analysis models struggle with sarcastic expressions, while many sarcasm detection studies focus solely on classification without examining their impact on downstream sentiment and emotion interpretation.

Furthermore, dataset limitations and generalisation challenges continue to affect model performance. Many existing studies rely on domain-specific datasets or distant supervision techniques, which may not accurately reflect real-world scenarios. Additionally, while recent research has explored LLM-based approaches, there is still limited work on integrated sarcasm detection, sentiment correction, and emotion classification within a unified framework.

Therefore, this research addresses an important gap by investigating how large language models can be used to improve the interpretation of sarcastic product reviews. By combining sarcasm detection with sentiment and emotion analysis, this study aims to provide a more accurate and context-aware understanding of customer feedback in e-commerce environments.

5 Chapter 3: METHODOLOGY

5.1 Overview and System Workflow

This chapter presents the methodology adopted in this study, including system design, data preparation, preprocessing techniques, model selection, evaluation strategy, and implementation environment. The aim of this research is to analyse sentiment and sarcasm in product reviews and to investigate how different models interpret contextual meaning.

The overall system follows a structured natural language processing pipeline consisting of two main tasks: sentiment analysis and sarcasm detection. Sentiment analysis focuses on classifying reviews into positive, neutral, or negative categories, while sarcasm detection identifies whether a review is sarcastic or standard. A comparative approach was adopted to evaluate the performance of traditional machine learning models, transformer-based models, and large language models.

The workflow begins with raw textual data, which is cleaned and preprocessed before being passed through different models for classification. The outputs of these models are then evaluated using multiple performance metrics and further analysed using custom examples. A conceptual representation of this pipeline includes preprocessing, feature representation, model inference, and evaluation stages. This structure ensures consistency across all experiments and allows fair comparison between models.

5.2 Data Acquisition and Preparation

5.2.1 Data Collection

The dataset used in this study was obtained from Kaggle and consists of Amazon product reviews. The original dataset contained multiple columns; however, only the relevant attributes were retained. Specifically, the dataset was reduced to the review text and rating columns. This simplification allowed the analysis to focus entirely on textual content while enabling efficient processing and labelling.

5.2.2 Data Cleaning and Label Preparation

The dataset was cleaned by removing unnecessary columns and handling missing values. The rating column was then used to assign sentiment labels. Reviews with ratings of 4 and 5 were labelled as positive, rating 3 as neutral, and ratings 1 and 2 as negative. This mapping enabled the transformation of numerical ratings into categorical labels suitable for supervised learning.

5.2.3 Sarcasm Dataset Construction

In addition to the sentiment dataset, a separate dataset was constructed specifically for sarcasm detection. This dataset consisted of two distinct types of data:

- Real Amazon product reviews, which were treated as non-sarcastic (standard)
- Artificially generated reviews, which were labelled as sarcastic

The sarcastic reviews were generated using varied sentence structures, tone, and phrasing to avoid repetitive patterns. The aim was to simulate realistic sarcastic expressions rather than relying on fixed templates. This distinction between real and generated data was necessary to create a labelled dataset for sarcasm detection, as real-world sarcasm annotations are often limited.

5.2.4 Data Leakage Control

To ensure the reliability of the experimental results, several checks were performed to prevent data leakage. First, the dataset was examined to ensure that no identical or near-duplicate reviews appeared in both training and testing sets. Second, the text data was inspected to confirm that no explicit label indicators, such as the words “sarcastic” or “positive”, were present within the reviews. Finally, the generated sarcastic reviews were reviewed to minimise repetitive phrasing, reducing the risk of models memorising patterns instead of learning meaningful features.

5.2.5 Train-Test Split

The dataset was divided into training and testing sets using an 80:20 split. Eighty percent of the data was used for training, while the remaining twenty percent was reserved for testing. Stratified sampling was applied to ensure that each class was proportionally represented in both sets. This approach ensures a fair evaluation of model performance and reduces bias.

5.3 Data Preprocessing

Before training the models, the text data was preprocessed to improve consistency and performance. All text was converted to lowercase, and punctuation and special characters were removed. For the SVM model, stopwords were removed using the NLTK library to reduce noise in the data. Tokenisation was applied to break text into smaller units suitable for analysis.

For traditional machine learning, TF-IDF vectorisation was used to convert text into numerical features by capturing the importance of words within the dataset. For transformer-based models such as BERT, tokenisation was handled using a pre-trained tokenizer, which preserves contextual relationships between words.

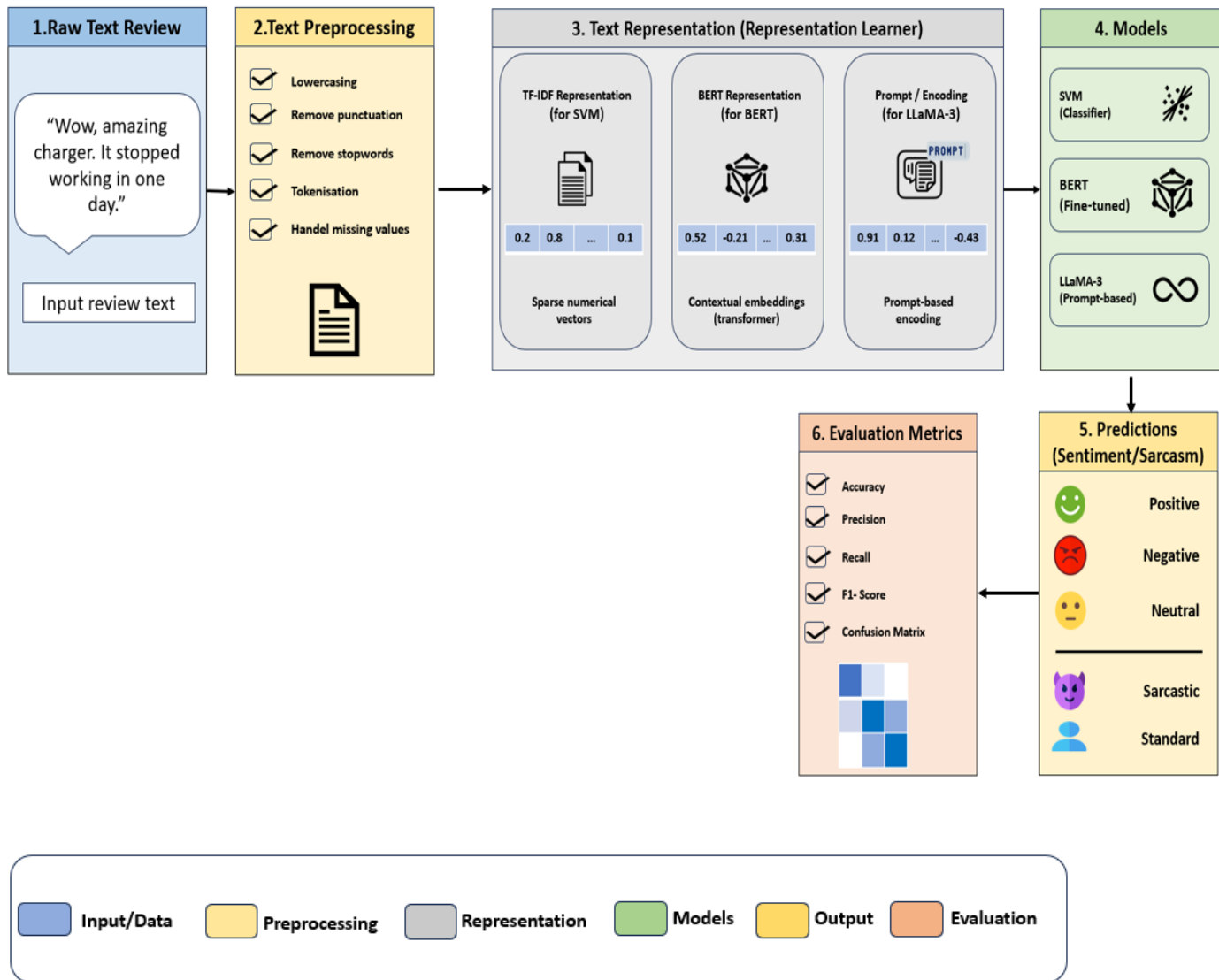


Fig 5.1 NLP pipeline showing preprocessing, representation learning and model classification.

5.4 Model Selection Rationale

Different models were selected to evaluate performance across varying levels of complexity and contextual understanding. The Support Vector Machine (SVM) was used as a baseline model due to its effectiveness in text classification and its ability to provide interpretable feature importance.

BERT was selected for fine-tuning because of its strong capability in capturing contextual meaning through bidirectional encoding. In addition, a zero-shot BERT model was used to assess the ability of pre-trained models to perform classification without task-specific training.

Finally, LLaMA-3 was used as a large language model capable of performing classification through prompt-based inference. This model was included to evaluate how well modern LLMs can understand implicit meaning and sarcasm without relying heavily on labelled data.

5.5 Evaluation Metrics

To evaluate model performance, multiple metrics were used rather than relying solely on accuracy. While accuracy provides an overall measure of correctness, it can be misleading in cases where class distributions are imbalanced or where subtle linguistic features, such as sarcasm, are involved.

Precision was used to measure how many predicted positive instances were actually correct. This is particularly important in sarcasm detection, where false positives may lead to incorrect interpretation of user intent. Recall was used to measure how many actual positive instances were correctly identified, ensuring that the model does not miss important cases of sarcasm or negative sentiment.

The F1-score was used as a balanced metric that combines precision and recall, making it suitable for evaluating performance in tasks where both false positives and false negatives are important. In addition, confusion matrices were used to visualise model performance and identify patterns of misclassification.

This combination of metrics provides a more comprehensive evaluation and ensures that the models are assessed not only on accuracy but also on their ability to generalise and handle complex language patterns.

In addition to these metrics, further analysis of model behaviour was performed by examining feature importance, particularly for the SVM model, to better understand how predictions were made.

5.6 Environment, Tools, and Libraries

The implementation of this project was carried out using Python as the primary programming language. Google Colab was used as the development environment due to its cloud-based GPU support, which enabled efficient model training.

Scikit-learn was used for implementing the SVM model and evaluation metrics, while the Hugging Face Transformers library was used for BERT and zero-shot models. PyTorch was used for training deep learning models, and NLTK was used for text preprocessing. Matplotlib was used for visualising results such as word frequency distributions.

5.7 Validation Strategy

To ensure reliable results, both quantitative and qualitative validation methods were used. Quantitative validation involved evaluating the models on the test dataset using standard metrics. Qualitative validation involved testing the models on custom-written examples to assess their ability to handle subtle sarcasm and ambiguous sentiment.

This combined approach ensures that the models are not only accurate on structured datasets but also capable of generalising to real-world scenarios.

6 CHAPTER 4: IMPLEMENTATION

6.1 Overview and System Workflow

This chapter presents the implementation of the proposed system and the results obtained from different models. The study followed a structured workflow, beginning with data preprocessing and ending with evaluation and analysis of model performance.

The implementation was carried out in stages, including dataset preparation, preprocessing, model training, and evaluation. Multiple models were applied, including SVM, fine-tuned BERT, zero-shot BERT, and LLaMA-3. The purpose of this approach was to compare traditional machine learning methods with modern transformer-based and large language models.

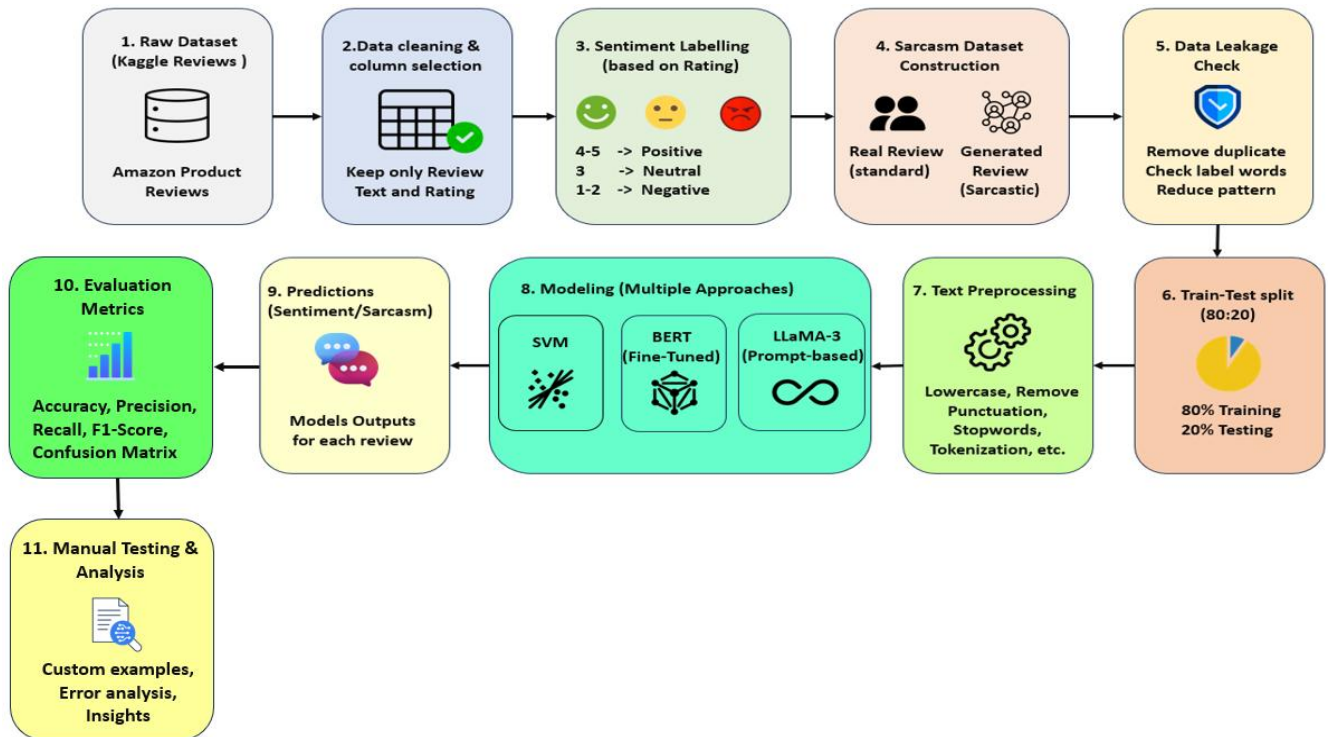


Fig 6.1: Complete workflow of sentiment analysis and sarcasm detection system

The above figure illustrates the complete workflow followed in this study. The process begins with raw product review data, which is cleaned and transformed into a structured format. Sentiment labels are assigned based on review ratings, and a separate sarcasm dataset is constructed using both real and generated reviews.

The data is then preprocessed and passed through different models for classification. Finally, predictions are evaluated using multiple performance metrics and further analysed using custom examples. This structured workflow ensures consistency across all experiments and allows fair comparison between models.

6.2 Data Preparation

The dataset used in this study was obtained from Kaggle and consists of Amazon product reviews. The original dataset contained several attributes; however, only the review text and rating columns were retained, as they were most relevant for the tasks. For sentiment analysis, the rating values were

converted into categorical labels, where ratings 4 and 5 were labelled as positive, rating 3 as neutral, and ratings 1 and 2 as negative. For sarcasm detection, a separate dataset was created in which real product reviews were treated as standard (non-sarcastic), while additional reviews were generated and labelled as sarcastic. This approach allowed the creation of a balanced dataset for training and testing. The dataset was then divided into training and testing sets using an 80:20 split, and stratified sampling was applied to ensure that each class was proportionally represented.

6.3 Data Preprocessing

The text data was preprocessed to improve consistency and prepare it for model training. All reviews were converted to lowercase to ensure uniformity, and punctuation and special characters were removed to reduce noise in the dataset. Tokenisation was applied to break the text into smaller units suitable for analysis. For the SVM model, additional preprocessing steps such as stopwords removal were performed using the NLTK library to eliminate commonly occurring words that do not contribute significantly to meaning. The processed text was then converted into numerical form using TF-IDF vectorisation, which captures the importance of words within the dataset. For transformer-based models such as BERT, pre-trained tokenizers were used to preserve contextual relationships between words and generate suitable input representations for the model.

6.4 Model Implementation

6.4.1 Support Vector Machine (SVM)

The SVM model was implemented using the Scikit-learn library. TF-IDF features were used to represent the text data numerically. The model was trained on the training dataset and then used to generate predictions on the test set. Predictions were also generated on custom examples to observe how the model behaves on unseen inputs.

SVM was selected as a baseline model due to its effectiveness in traditional text classification tasks and its ability to perform well on smaller datasets. It also provides interpretable results, making it useful for understanding how individual features contribute to predictions.

6.4.2 Fine-Tuned BERT Model

The BERT model was implemented using the Hugging Face Transformers library. Specifically, the bert-base-uncased model was used as the base architecture. This model was selected because it is a widely used pre-trained transformer model with a good balance between performance and computational efficiency, making it suitable for medium-sized datasets such as the one used in this study. A pre-trained BERT model was fine-tuned on the dataset for both sentiment analysis and sarcasm detection tasks. The model was trained using labelled data and evaluated on the test set. Predictions were also generated on custom examples to verify model behaviour.

During fine-tuning, the model was adapted to the classification task by adding a sequence classification layer on top of the pre-trained encoder. The input text was tokenised using the corresponding BERT tokenizer, and attention masks were used to handle variable-length inputs. The model was trained for multiple epochs using a standard optimisation approach, allowing it to adjust its internal weights based

on the dataset. This process enables the model to learn task-specific patterns while retaining its general language understanding.

BERT was chosen due to its strong ability to capture contextual relationships in text through bidirectional encoding. Unlike traditional models, BERT considers the full context of a sentence, making it particularly suitable for handling complex language patterns such as sarcasm and implicit sentiment. Among the available transformer models, BERT provides a strong baseline for fine-tuning and has been widely adopted in sentiment analysis and sarcasm detection tasks in recent research.

6.4.3 Zero-Shot BERT

The zero-shot BERT model was implemented using a pre-trained transformer model available through the Hugging Face pipeline, specifically designed for natural language inference tasks. In this approach, no additional training was performed on the dataset. Instead, the model was provided with candidate labels such as “positive”, “neutral”, and “negative” for sentiment analysis, and “sarcastic” and “standard” for sarcasm detection.

The model evaluates the relationship between the input text and the candidate labels using its pre-trained knowledge and assigns probabilities to each label. This allows the model to perform classification without requiring task-specific training data. The zero-shot approach was included to assess how well a general-purpose language model can adapt to new tasks and datasets without fine-tuning.

6.4.4 LLaMA-3 Model

The LLaMA-3 model was used through prompt-based inference. Specifically, the Meta LLaMA-3 8B model was used, which is an open-source large language model developed by Meta and contains approximately 8 billion parameters. Instead of training, the model was given structured prompts instructing it to classify sentiment and sarcasm in product reviews. Predictions were generated based on the prompts provided to the model.

LLaMA-3 was included as a large language model to evaluate its ability to understand implicit meaning and contextual nuances through natural language prompts. Being an open-source model, it allows flexibility in experimentation and does not require proprietary access. Compared to traditional and transformer-based models, LLaMA-3 relies more on general language understanding rather than task-specific training. Its large number of parameters enables it to capture complex semantic relationships, making it suitable for tasks such as sarcasm detection, where meaning is often implied rather than explicitly stated.

The 8 billion parameter variant was selected as it provides a balance between performance and computational efficiency. While larger versions of the model are available, they require significantly more computational resources and are not always practical to run in environments such as Google Colab. Given the relatively small dataset used in this study, the 8B model was sufficient to capture meaningful patterns without introducing unnecessary computational overhead.

The use of LLaMA-3 in this project provides insight into how modern large language models perform on relatively small datasets without fine-tuning, highlighting their potential for generalisation and contextual reasoning.

6.5 Implementation Environment

The implementation of this project was carried out using Python within the Google Colab environment. Google Colab was selected due to its ease of use and its ability to provide cloud-based computational resources, including access to GPUs, which are particularly useful for training transformer-based models such as BERT. This environment also allowed seamless integration with external datasets and libraries, making the overall development process more efficient.

A range of libraries and frameworks were used to support different stages of the implementation. Scikit-learn was used for building and training the SVM model, as it provides efficient tools for traditional machine learning tasks. The Hugging Face Transformers library was used to implement and fine-tune the BERT model, offering access to pre-trained models and tokenisers. PyTorch was used as the underlying deep learning framework for handling model training and tensor operations. For text preprocessing, the Natural Language Toolkit (NLTK) library was used to perform tasks such as stopword removal and tokenisation. In addition, Matplotlib was used to generate visualisations, particularly for analysing word distributions and presenting results.

6.6 Summary

This chapter described the implementation process of the proposed system, including data preparation, preprocessing, and model development. Different models were applied to perform sentiment analysis and sarcasm detection. The results obtained from these models are presented and analysed in the following chapters.

7 CHAPTER 5: TESTING AND RESULTS

7.1 Overview

This chapter presents the results obtained from the implementation of different models for sentiment analysis and sarcasm detection. The performance of each model was evaluated using standard classification metrics, and predictions were also generated on custom examples to observe behaviour on unseen inputs.

7.2 Evaluation Metrics

The performance of the models was evaluated using accuracy, precision, recall, and F1-score. Accuracy provides an overall measure of correct predictions, while precision and recall give insight into how well each class is identified. The F1-score was used as a balanced metric that considers both precision and recall. Confusion matrices were also used to visualise classification performance.

7.3 SVM Results

The SVM model was evaluated on both sentiment analysis and sarcasm detection tasks. For sentiment analysis, the model achieved an accuracy of approximately 88.5%, with strong performance across all classes. For sarcasm detection, the model achieved an accuracy of approximately 99% on the dataset.

In addition to evaluation metrics, predictions were generated on custom examples to observe the behaviour of the model on unseen inputs.

7.4 Fine-Tuned BERT Results

The fine-tuned BERT model was applied to both sentiment analysis and sarcasm detection tasks. For sentiment analysis, the model achieved an accuracy of approximately 92%, demonstrating strong performance across the dataset. For sarcasm detection, the model achieved an accuracy of approximately 99%. Predictions were also generated on custom examples to further observe the model's output.

7.5 Zero-Shot BERT Results

The zero-shot BERT model was evaluated without additional training on the dataset. For sentiment analysis, the model achieved an accuracy of approximately 50%, while for sarcasm detection it achieved approximately 61% accuracy. Predictions were generated based on the predefined labels, allowing the model to classify text without task-specific training.

7.6 LLaMA-3 Results

The LLaMA-3 model was evaluated using prompt-based inference for both sentiment analysis and sarcasm detection tasks. For sentiment analysis, the model achieved an accuracy of approximately 49%, while for sarcasm detection it achieved approximately 70% accuracy. Predictions were generated based on structured prompts provided to the model.

7.7 Summary

This chapter presented the results obtained from different models for sentiment analysis and sarcasm detection. The results show variation in performance across models, with differences observed between traditional, transformer-based, and large language model approaches. A detailed discussion and analysis of these findings is provided in the next chapter.

8 CHAPTER 6: EVALUATION AND DISCUSSION

8.1 Overview

This chapter presents a detailed evaluation and discussion of the results obtained from the implemented models. While the previous chapter focused on reporting performance metrics, this chapter aims to interpret those results, analyse model behaviour, and identify key strengths and limitations of each approach.

8.2 Evaluation of SVM Model

The SVM model achieved strong performance on both sentiment analysis and sarcasm detection tasks, particularly in terms of overall accuracy. However, further evaluation revealed that the model's performance was largely influenced by lexical patterns present in the dataset rather than a deeper understanding of context. The model struggled with neutral reviews, often misclassifying them as either positive or negative. This is expected, as neutral sentiment is typically less distinct and more dependent on contextual interpretation.

```
Custom review predictions:
Review: This product is excellent and works perfectly. I am very happy with it.
Predicted sentiment: positive
Decision scores: [-0.47755511 -0.89235822 0.37460057]

Review: The product is okay, nothing special.
Predicted sentiment: negative
Decision scores: [-0.28998532 -0.41400445 -0.29194995]

Review: Very disappointing item. It stopped working after two days.
Predicted sentiment: neutral
Decision scores: [-0.50688095 0.23399515 -0.72510694]
```

Fig 8.1: SVM predictions on sentiment examples

```
Custom review predictions:
Review: This product is fantastic and works perfectly.
Predicted style: standard
Decision score: 0.6777129249934591

Review: dissappointing charger. I thought it would stop working in one day, but is working absolutely fine.
Predicted style: standard
Decision score: 0.1766643655159999

Review: Absolutely brilliant. I love how it stopped working immediately.
Predicted style: sarcastic
Decision score: -0.20576669633150435

Review: The product is average, not too bad and not too good.
Predicted style: standard
Decision score: 0.5134961673720816
```

Fig 8.2: SVM predictions on sarcasm examples

As shown in the figures, when tested on unseen examples, the model failed to correctly identify subtle sarcasm and often classified such reviews as standard. This suggests that the model may have learned dataset-specific patterns rather than genuinely understanding sarcasm. While the model performs well on clearly defined examples, it struggles with more subtle or ambiguous cases. For instance, reviews expressing mild sentiment or implicit meaning are sometimes misclassified. Overall, this indicates that the SVM model relies heavily on surface-level word patterns rather than deeper contextual understanding.

8.3 Word Distribution and Feature Importance Analysis

To further understand the behaviour of the SVM model, word frequency analysis and feature importance were examined.

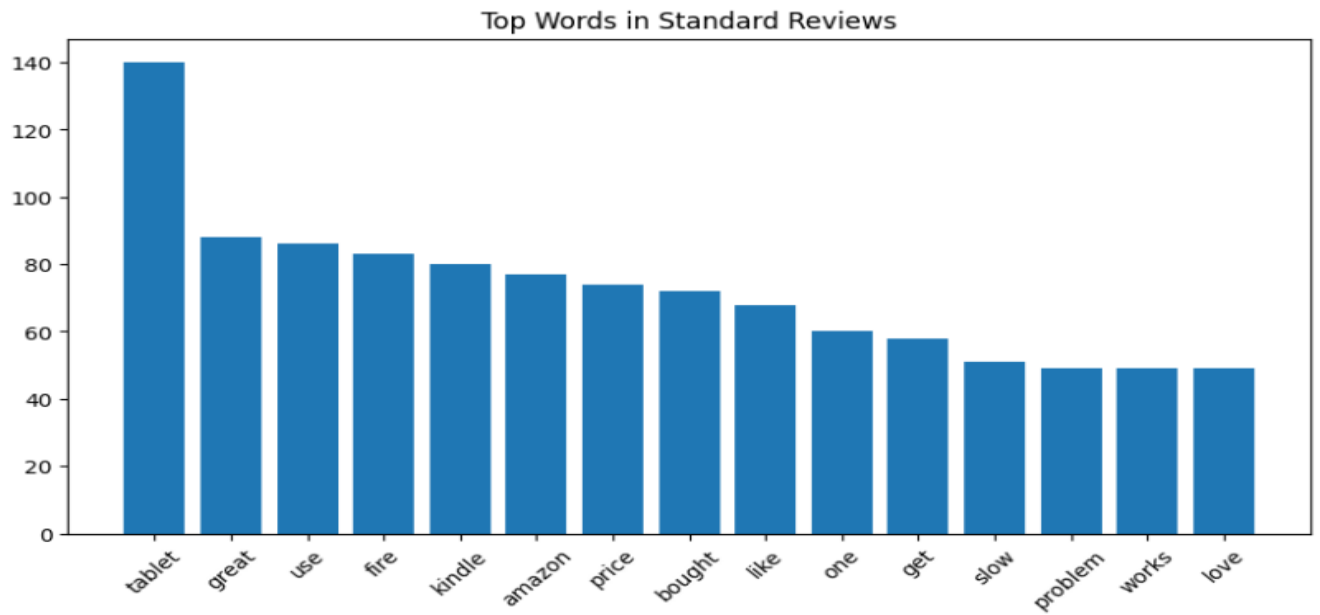


Fig 8.3: Most frequent words in standard reviews

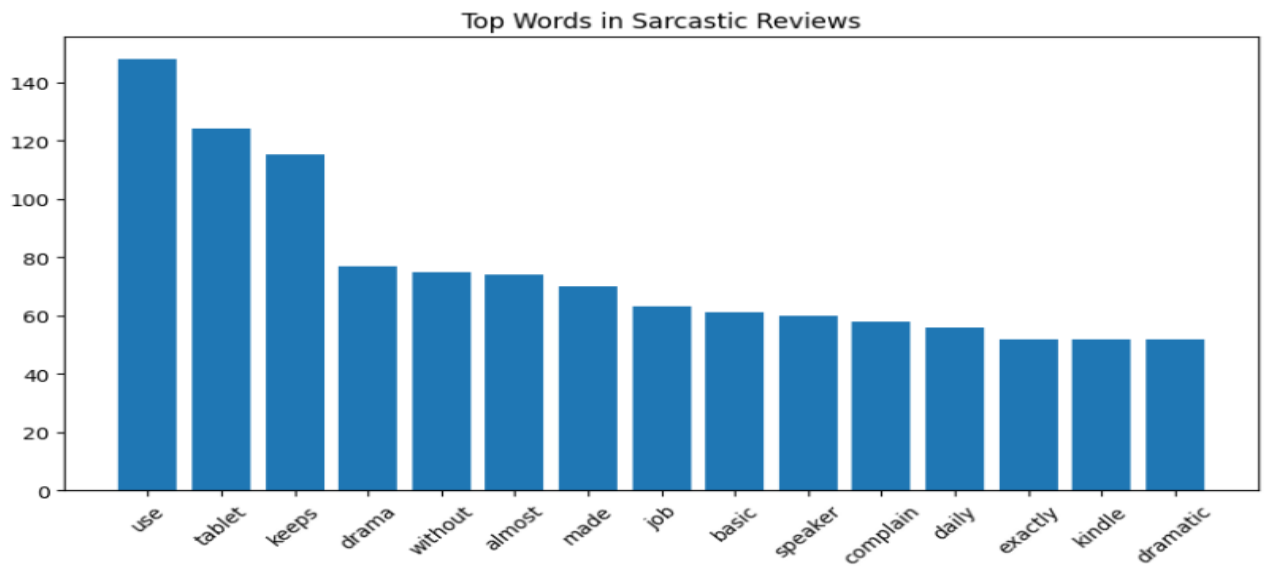


Fig 8.4: Most frequent words in sarcastic reviews

Top STANDARD words (SVM):	Top SARCASTIC words (SVM):
great 0.787881299853121	exactly -0.9713228987333306
amazon 0.7807827364041409	keeps -0.9300253753559761
problem 0.6398518922376831	just love -0.9058423751201975
bought 0.5850715720642924	dramatic -0.8941742844899824
price 0.5686902471140303	average -0.8455573191027195
loves 0.5438718589529956	job -0.8152747991623248
bad 0.495336814396546	felt -0.7804998687730228
tv 0.4871588495515226	drama -0.7730182769837398
alexa 0.48048755990497044	memorable -0.7444469633398662
purchased 0.46638985152000795	complain -0.7406621376990508
apps 0.4659319576443192	functionality -0.7383261312065611
kids 0.45906085485063386	way -0.7339491133491499
slow 0.45210013955673206	just -0.7244736634034826
reading 0.4501396687673855	actually -0.7025941311966214
nice 0.43645550948327405	honestly -0.6961000011871922

Fig 8.5: Important features learned by SVM model

The results show that standard reviews are dominated by product-related words such as “great”, “price”, and “works”, while sarcastic reviews contain more expressive or exaggerated terms such as “dramatic”, “exactly”, and “complain”.

Further analysis of the SVM feature weights revealed that the model assigns high importance to these same words when making predictions. Words such as “great” and “nice” are strongly associated with standard reviews, while words such as “dramatic” and phrases such as “just love” are associated with sarcastic reviews.

This indicates that the model relies heavily on lexical patterns rather than understanding contextual meaning. As a result, although the model achieves very high accuracy on the dataset, it struggles to generalise when faced with more subtle or complex examples of sarcasm.

8.4 Evaluation of Fine-Tuned BERT Model

The fine-tuned BERT model demonstrated strong performance across both sentiment analysis and sarcasm detection tasks, primarily due to its ability to capture contextual relationships within text. Compared to the SVM model, BERT showed improved handling of complex sentence structures and implicit meaning, resulting in more accurate predictions on the dataset.

Review: This product is excellent and works perfectly. I am very happy with it.
Predicted sentiment: positive

Review: The item is okay. Not bad, but nothing special.
Predicted sentiment: negative

Review: Very disappointing product. It stopped working after two days.
Predicted sentiment: negative

Fig 8.6: Fine-tuned BERT predictions for sentiment analysis

Review: This product is fantastic and works perfectly.
Predicted: standard

Review: Wow, amazing charger. It stopped working in one day.
Predicted: standard

Review: Absolutely brilliant. I love how it stopped working immediately.
Predicted: sarcastic

Review: The product is average, not too bad and not too good.
Predicted: standard

Review: disappointing charger. I thought it would stop working in one day, but is working absolutely fine.
Predicted: standard

Fig 8.7: Fine-tuned BERT predictions for sarcasm detection

However, manual testing revealed that the model still struggled with subtle sarcasm and borderline sentiment expressions. For instance, reviews with implicit or mild sentiment, such as *“The item is okay. Not bad, but nothing special”*, were sometimes misclassified, indicating difficulty in distinguishing between neutral and negative sentiment.

Similarly, sarcasm that relies on implicit contradiction rather than explicit wording was not always correctly identified. This suggests that although BERT provides improved contextual understanding compared to traditional models, it may still depend on patterns present in the training data and may not fully capture deeper pragmatic meaning in all cases.

8.5 Evaluation of Zero-Shot BERT

The zero-shot BERT model showed moderate performance, with lower accuracy compared to the fine-tuned model.

Review: This product is fantastic and works perfectly.
Prediction: standard
Scores: {'standard': 0.9846423864364624, 'sarcastic': 0.015357679687440395}

Review: Wow, amazing charger. It stopped working in one day.
Prediction: sarcastic
Scores: {'sarcastic': 0.852803111076355, 'standard': 0.14719687402248383}

Review: Absolutely brilliant. I love how it stopped working immediately.
Prediction: sarcastic
Scores: {'sarcastic': 0.8153532147407532, 'standard': 0.18464677035808563}

Review: The product is average, not too bad and not too good.
Prediction: standard
Scores: {'standard': 0.9915837645530701, 'sarcastic': 0.00841626524925232}

Review: disappointing charger. I thought it would stop working in one day, but is working absolutely fine.
Prediction: sarcastic
Scores: {'sarcastic': 0.9321328997612, 'standard': 0.06786715984344482}

Fig 8.8: Zero-shot BERT predictions for sarcasm detection

```

Review: This product is excellent and works perfectly.
Prediction: positive
Scores: {'positive': 0.9958714842796326, 'neutral': 0.0028274324722588062, 'negative': 0.001301125972531736}
-----
Review: The product is okay, nothing special.
Prediction: neutral
Scores: {'neutral': 0.9695791006088257, 'positive': 0.018806379288434982, 'negative': 0.01161450520157814}
-----
Review: Very disappointing item.
Prediction: negative
Scores: {'negative': 0.9982876181602478, 'positive': 0.000954939576331526, 'neutral': 0.000757489528041333}
-----

```

Fig 8.9: Zero-shot BERT predictions for sentiment analysis

Despite lower overall accuracy, the zero-shot BERT model showed good generalisation on unseen examples. As seen in the figures, it was able to correctly identify sarcasm in reviews where there was a clear contradiction, such as positive words being used to describe a negative experience. This shows that the model can understand the general meaning of a sentence even without being trained on the dataset.

However, the model did not perform well across the full dataset. While it handled clear examples correctly, it was not consistent when dealing with more unclear or mixed reviews. In particular, it had difficulty in correctly classifying sentiment in some cases, especially when the difference between neutral and negative was not very obvious. This shows that zero-shot models can understand general language, but they may struggle when applied to a specific dataset without training.

8.6 Evaluation of LLaMA-3 Model

The LLaMA-3 model showed different behaviour compared to both traditional and transformer-based models.

```

Review: This product is fantastic and works perfectly.
Predicted style: standard

Review: Wow, amazing charger. It stopped working in one day.
Predicted style: sarcastic

Review: Absolutely brilliant. I love how it stopped working immediately.
Predicted style: sarcastic
-----
Review: The product is average, not too bad and not too good.
Predicted style: standard

Review: disappointing charger. I thought it would stop working in one day, but is working absolutely fine.
Predicted style: sarcastic
-----

```

Fig 8.10: LLaMA-3 predictions for sentiment analysis

Review: This product is excellent and works perfectly. I am very happy with it.
Predicted sentiment: positive

Review: The item is okay. Not bad, but nothing special.
Predicted sentiment: neutral

Review: Very disappointing product. It stopped working after two days.
Predicted sentiment: negative

Fig 8.11: LLaMA-3 predictions for sarcasm detection

The model demonstrated a strong ability to detect sarcasm, particularly in cases involving contradiction or exaggerated expressions. However, it tended to over-predict sarcasm in some cases, indicating a bias towards interpreting ambiguous language as sarcastic.

For sentiment analysis, the model showed weaker performance, particularly in distinguishing between neutral and negative sentiment. This may be due to the lack of task-specific fine-tuning and reliance on general language understanding.

8.7 Comparative Discussion

A comparison of all models reveals clear differences in performance and behaviour across both sentiment analysis and sarcasm detection tasks. The SVM model achieved high accuracy, particularly in sarcasm detection, but its performance is largely driven by surface-level word patterns rather than true contextual understanding. This makes it effective on structured data but less reliable when dealing with subtle or implicit language. The fine-tuned BERT model provided the strongest overall performance, achieving high accuracy in both tasks while demonstrating a better ability to capture contextual relationships within text. However, it still showed some limitations when handling borderline sentiment and subtle sarcasm, suggesting a degree of dependence on patterns learned during training. The zero-shot BERT model, while achieving lower accuracy, demonstrated stronger generalisation capabilities, as it was able to classify unseen examples without task-specific training. This indicates its flexibility, although it lacks precision when applied to domain-specific datasets. The LLaMA-3 model showed a different behaviour compared to other models, with relatively lower accuracy in sentiment analysis but improved performance in sarcasm detection. Its ability to capture implicit meaning and contextual nuances highlights the strength of large language models, although its predictions were sometimes inconsistent due to the absence of fine-tuning.

To provide a clearer comparison of model performance, a summary is presented in Table 6.1.

Model	Sentiment Accuracy	Sarcasm Accuracy	Generalisation
SVM	89%	99%	Poor
BERT(Fine-tuned)	92%	99%	Moderate
BERT(Zero-shot)	50%	61%	Good
LLaMA-3	49%	70%	Strong

Table 8.1: Comparison of model performance across tasks

The comparison shows that high accuracy does not always indicate better understanding. While SVM and fine-tuned BERT achieved very high accuracy, their ability to generalise to unseen or more complex examples is limited. In contrast, the zero-shot BERT and LLaMA-3 models demonstrated stronger

generalisation, particularly in handling unseen data and implicit meaning, despite having lower overall accuracy. This highlights an important trade-off between accuracy and generalisation when selecting models for real-world applications.

8.8 Limitations of the Study

While the results of this study are encouraging, there are several limitations that should be considered.

One of the main limitations is the dataset used in this project. The sarcasm dataset includes synthetically generated examples, which may not fully reflect how sarcasm is expressed in real-world reviews. As a result, the models may perform well on the dataset but struggle with more natural and complex user-generated content.

Another limitation is the size of the dataset. A relatively small dataset was used, which may limit the ability of the models to generalise across different writing styles and domains.

In addition, sentiment analysis and sarcasm detection were treated as separate tasks, whereas in real-world scenarios they are closely related. This may affect the accuracy of interpreting user intent.

Finally, large language models such as LLaMA-3 and zero-shot BERT were used without fine-tuning. While this helps evaluate generalisation, it may also lead to inconsistent performance.

These limitations highlight areas for improvement in future work.

8.9 Summary

This chapter presented the implementation and results of different models applied to sentiment analysis and sarcasm detection. The findings highlight the limitations of traditional models and the potential of large language models in understanding complex language patterns.

9 CHAPTER 7: CONCLUSION, REFLECTION AND FUTURE WORK

9.1 Conclusion

This study explored the problem of sentiment analysis and sarcasm detection in online product reviews using a range of machine learning, transformer-based, and large language models. The aim was to understand how different approaches perform when dealing with both direct sentiment and more complex language patterns such as sarcasm.

The results showed that traditional models such as SVM can achieve high accuracy, especially on structured datasets. However, their performance is largely based on recognising common word patterns rather than understanding the actual meaning of the text. This limits their ability to handle subtle or implicit language.

The fine-tuned BERT model provided the best overall performance across both sentiment analysis and sarcasm detection tasks. Its ability to consider the full context of a sentence allowed it to perform

better than traditional models, especially when dealing with more complex reviews. However, it still showed some difficulty when handling neutral sentiment and subtle sarcasm.

The zero-shot BERT model demonstrated a different behaviour. Although its overall accuracy was lower, it showed better generalisation on unseen examples. This highlights its ability to apply general language knowledge without task-specific training, although it lacks consistency when used on a specific dataset.

The LLaMA-3 model showed strong capability in detecting sarcasm, particularly in cases where there was a clear contradiction in the sentence. However, its performance on sentiment analysis was weaker, and its predictions were sometimes inconsistent. This reflects the nature of large language models, which rely on general understanding rather than being trained for a specific task.

Overall, the study highlights that no single model performs perfectly across all aspects of sentiment and sarcasm detection. There is a clear trade-off between accuracy and generalisation. While fine-tuned models perform well on structured tasks, large language models offer flexibility and better handling of implicit meaning.

9.2 Reflection:

This project highlights that high accuracy does not always mean better understanding. For example, the SVM model achieved very high accuracy but relied heavily on word patterns rather than true contextual meaning. This was an important learning point, as it showed traditional model can appear effective while still lacking understanding. Another key insight was that trade-off between accuracy and generalisation. While fine-tuned models such as BERT performed best on the dataset, they struggled with more subtle and unseen examples. In Contrast, zero-shot and large language models showed better generalisation but lower consistency. This demonstrates that selecting the right model depends on real-world application and not just performance metrics. Overall, this project improved my understanding of how different NLP models behave and highlighted the challenges of working with complex language such as sarcasm.

9.3 Future Work

While this study provides useful insights, there are several areas that can be explored in future work to improve the results further.

One important direction is to combine sarcasm detection with sentiment classification. Instead of treating them as separate tasks, future work could focus on identifying whether a sarcastic review expresses positive, negative, or neutral sentiment. This would provide a more accurate understanding of user opinions.

Another area for improvement is emotion analysis. Reviews often contain emotions such as frustration, disappointment, or satisfaction, which are not fully captured by simple sentiment labels. Extending this work to include emotion detection could provide deeper insights into customer feedback.

In addition, the use of larger and more diverse datasets could improve model performance and generalisation. The current dataset is relatively small, and increasing its size could help models learn more varied language patterns.

A key improvement would be to use real-world sarcastic reviews collected from actual users instead of relying only on synthetic or pre-annotated datasets. Real user-generated data often contains more

natural and complex expressions of sarcasm, which can help models learn more realistic patterns and improve performance in practical applications.

Further improvements could also be made by applying advanced prompting techniques for large language models, such as few-shot learning or chain-of-thought prompting. This may help improve consistency and accuracy, particularly in sarcasm detection.

Finally, future work could explore combining different models into a single system. For example, traditional models could be used for efficient classification, while large language models could handle more complex cases. Such hybrid approaches may provide a better balance between accuracy, efficiency, and generalisation.

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